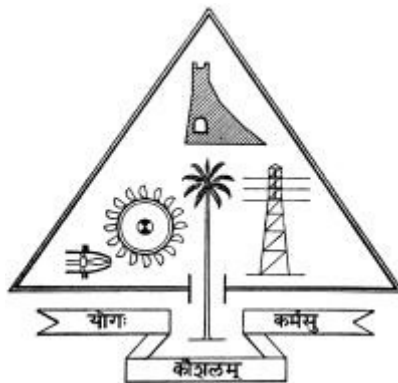


SMARTEDIT: A HUMAN-IN-THE-LOOP AI FRAMEWORK FOR MULTI-FOOTAGE VIDEO EDITING

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ABSTRACT

Video editing is a time-consuming and skill-intensive process that requires reviewing raw footage, identifying usable clips, removing defective or redundant content, and arranging clips into a coherent narrative. These repetitive tasks increase editing time, particularly when working with large volumes of multi-footage recordings. This project proposes **SmartEdit**, a Human-in-the-Loop AI framework for multi-footage video editing that combines natural language understanding, multi-modal content analysis, and human creative supervision to automate repetitive editing tasks while ensuring that the final creative decisions remain with the editor.

The proposed system allows users to import raw video clips and describe their editing requirements using natural language prompts, such as *"remove shaky footage, trim silence, and edit it like a podcast."* A Large Language Model (LLM) interprets the user's editing intent and identifies the desired editing style. Simultaneously, each imported clip is processed by a multi-modal Clip Analysis Engine that uses OpenCV for motion and shake detection, YOLO for object and person detection, CLIP for scene understanding, Whisper for speech transcription, and Librosa for silence and audio quality analysis. These AI models convert raw video into structured metadata describing the content, quality, and context of every clip.

Using the generated metadata together with the user's editing intent, the LLM produces intelligent editing recommendations, including detection of shaky or blurry footage, silence trimming, duplicate clip identification, speaking segment detection, and optimal trim-point suggestions. Unlike fully automated video editing systems, the proposed framework follows a **Human-in-the-Loop** approach in which AI recommendations are

presented to the editor for review rather than applied permanently. Flagged clips are moved to a review track where the editor can accept, reject, modify, or restore every recommendation before proceeding with the edit. The system also performs content-aware editing by recommending suitable editing styles and timeline structures for different video categories such as podcasts, vlogs, travel videos, wedding videos, and cinematic productions.

After the AI generates an initial draft timeline, the human editor can further refine the project by rearranging clips, adjusting trim points, replacing selected footage, and adding transitions, effects, or music before exporting the final edited video. This collaborative workflow combines the efficiency of AI-assisted automation with human creativity and editorial control.

By integrating natural language understanding, multi-modal AI analysis, intelligent editing recommendations, and a transparent Human-in-the-Loop review process, the proposed system aims to reduce the time and manual effort required for video editing while maintaining flexibility, transparency, and complete human control over the final creative output.

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LIST OF ABBREVIATIONS

AI	– Artificial Intelligence
LLM	– Large Language Model
HITL	– Human-in-the-Loop
ML	– Machine Learning
DL	– Deep Learning
CNN	– Convolutional Neural Network
CV	– Computer Vision
NLP	– Natural Language Processing
OpenCV	– Open Source Computer Vision Library
YOLO	– You Only Look Once
CLIP	– Contrastive Language–Image Pre-training
GUI	– Graphical User Interface
API	– Application Programming Interface
JSON	– JavaScript Object Notation
FFmpeg	– Fast Forward Moving Picture Experts Group
Qt	– Qt Framework (Cross-Platform C++ Application Framework)
C++	– C Plus Plus (Programming Language)
CPU	– Central Processing Unit
GPU	– Graphics Processing Unit
FPS	– Frames Per Second

CHAPTER 1

INTRODUCTION

The rapid advancement of digital technology has transformed video into one of the most popular forms of communication, education, entertainment, marketing, and social media content creation. The widespread availability of smartphones, DSLR cameras, action cameras, and other recording devices has enabled users to capture high-quality videos with ease. As a result, content creators frequently record large amounts of raw footage for podcasts, vlogs, travel videos, interviews, educational content, and short films. Although capturing video has become simple, editing this footage into a polished final video remains a time-consuming and skill-intensive process. Editors must manually review every clip, remove unwanted sections, trim silence, identify poor-quality footage, organize clips, and construct a meaningful timeline before beginning the creative aspects of editing.

Professional video editing applications such as Adobe Premiere Pro, DaVinci Resolve, and CapCut provide powerful editing tools for trimming, sequencing, transitions, visual effects, color correction, and audio enhancement. While these applications offer extensive creative flexibility, they still rely heavily on manual workflows for repetitive tasks such as identifying shaky footage, removing silent sections, selecting usable clips, detecting duplicate recordings, and organizing the editing timeline. As the number of recorded clips increases, these repetitive operations consume a significant amount of time, reducing editing efficiency and increasing the workload of content creators.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML),

Computer Vision (CV), speech recognition, and Large Language Models (LLMs) have created new opportunities for intelligent video editing assistance. Computer Vision techniques can analyze video content to detect camera shake, scene changes, faces, objects, and motion, while speech processing models can identify spoken content and silent segments. Large Language Models enable users to communicate editing requirements using natural language, allowing editing software to better understand user intentions and automatically perform repetitive editing tasks based on simple text instructions.

Although several AI-powered editing tools have emerged in recent years, many of them focus on fully automated editing with limited user control. Such systems often make editing decisions without considering the user's creative preferences, resulting in outputs that may not match the desired storytelling style or editing objectives. Since video editing is a creative and subjective process, complete automation cannot replace the artistic judgment of human editors. Therefore, there is a growing need for intelligent editing systems that combine AI-driven automation with human creativity and decision-making.

Human-in-the-Loop (HITL) Artificial Intelligence provides an effective solution to this challenge. Rather than replacing the editor, HITL systems use Artificial Intelligence to analyze data, identify repetitive tasks, and generate intelligent recommendations while allowing the user to review and control every important decision. In the context of video editing, this approach enables AI to automate technical operations such as detecting shaky footage, trimming silence, identifying duplicate clips, analyzing video quality, and generating an initial timeline, while the editor retains complete authority to accept, reject, modify, or refine every AI-generated recommendation.

To address these challenges, this project proposes SmartEdit, a Human-in-the-Loop AI framework for intelligent video editing. The proposed system combines natural language understanding with multi-modal content analysis to simplify repetitive editing tasks. Users can import multiple video clips

and describe their editing requirements using natural language prompts such as "remove shaky footage, trim silence, and edit it like a podcast." A Large Language Model interprets the editing intent and determines the desired editing style, while a Clip Analysis Engine analyzes each video using OpenCV for motion analysis, YOLO for object and person detection, CLIP for scene understanding, Whisper for speech transcription, and Librosa for silence and audio analysis.

Based on the extracted metadata and the user's editing requirements, the system generates intelligent editing recommendations such as identifying shaky or blurry footage, detecting duplicate clips, removing silent sections, locating speaking segments, and suggesting suitable trim points. Instead of permanently applying these changes, the proposed framework follows a Human-in-the-Loop workflow in which all AI-generated recommendations are presented to the editor for review. Flagged clips are moved to a review track where the editor can accept, reject, modify, or restore every recommendation before continuing with the editing process.

Another important feature of SmartEdit is its content-aware editing assistance. The system can recommend suitable editing styles based on the user's prompt and the analyzed video content. For example, it can suggest silence removal for podcasts, fast-paced cuts for vlogs, smoother pacing for cinematic videos, or appropriate editing structures for travel videos and interviews. The AI generates an initial draft timeline that serves as a starting point rather than a completed edit, allowing the editor to make further creative adjustments.

After the AI generates the initial timeline, the editor can manually rearrange clips, adjust trim points, replace suggested footage, add transitions, apply visual effects, insert background music, and perform any additional creative modifications before exporting the final edited video. This collaborative workflow combines the speed and efficiency of Artificial Intelligence with the creativity, experience, and artistic judgment of human editors.

CHAPTER 2

LITERATURE REVIEW

The evolution of intelligent video editing systems remains a significant research area within multimedia computing, computer vision, and human-computer interaction ecosystems. Historically, video editing methodologies have been divided into two major approaches: traditional manual timeline-based editing systems and fully automated Artificial Intelligence-driven editing frameworks. Conventional professional editing platforms such as Adobe Premiere Pro, Final Cut Pro, and DaVinci Resolve provide extensive creative control through sophisticated editing interfaces; however, they require substantial human effort for footage organization, synchronization, scene selection, and timeline construction. Conversely, recent AI-powered editing systems attempt to automate these processes using computer vision and machine learning techniques. While these approaches demonstrate significant improvements in editing efficiency, both paradigms exhibit fundamental limitations when applied to the increasingly complex requirements of modern multi-camera video production environments.

Traditional video editing workflows rely heavily on manual inspection and decision-making performed by experienced editors. In multi-footage production scenarios, where recordings are captured simultaneously using smartphones, DSLR cameras, drones, and action cameras, editors must manually synchronize heterogeneous video streams, identify corresponding scenes, remove redundant footage, evaluate visual quality, and select appropriate camera perspectives. Although professional editing applications provide tools such as multi-camera synchronization and timeline management, these systems primarily function as productivity tools

rather than intelligent decision-support frameworks. Their dependence on manual operations creates significant bottlenecks when processing large volumes of footage, particularly for event-based productions such as weddings, sports coverage, interviews, and documentaries. Furthermore, traditional editing systems lack semantic understanding of video content and cannot independently determine which clips contain important events, which camera angle provides better storytelling value, or which sequences should be prioritized in the final composition.

To overcome the limitations of manual editing workflows, research has increasingly focused on automatic video editing and multi-camera content generation using computer vision techniques. Early automatic editing systems primarily investigated camera selection and shot transition strategies for structured environments such as sports broadcasting, television programs, and theatrical performances. These systems analyzed predefined characteristics such as speaker location, motion activity, scene boundaries, and camera viewpoints to determine suitable shots. Although these approaches established the foundation for automated camera selection, their effectiveness remained highly dependent on domain-specific assumptions. A camera selection strategy optimized for sports events, where actions follow predictable patterns, cannot be directly applied to informal environments such as wedding ceremonies or social media content creation, where emotional importance, human interaction, and storytelling context influence editing decisions. Additionally, many existing approaches focus on selecting individual shots rather than understanding the complete editing workflow, leaving challenges such as duplicate scene detection, footage ranking, and timeline generation unresolved.

With the advancement of Artificial Intelligence and deep learning architectures, video editing research has shifted from handcrafted rules toward data-driven approaches capable of learning complex visual patterns. Modern computer vision models, including convolutional neural networks, transformer-based architectures, and multimodal learning

systems, have demonstrated strong capabilities in tasks such as object detection, face recognition, activity recognition, scene classification, and video summarization. AI-assisted editing platforms have utilized these technologies to provide automated features including highlight generation, silence removal, subtitle creation, color enhancement, and content-based search. However, fully automated AI editing systems introduce a significant challenge: the reduction of human creative control. Video editing is not solely a technical process; it involves subjective decisions related to storytelling, emotional timing, visual composition, and artistic style. Existing AI models often operate as black-box systems, producing final outputs without providing sufficient explanation or allowing editors to modify intermediate decisions. This limitation reduces user trust and restricts adoption among professional editors who require precise control over the final production.

Parallel to AI-driven automation, Human-in-the-Loop (HITL) research has emerged as an important paradigm for designing collaborative intelligent systems. Instead of replacing human expertise, HITL approaches combine machine efficiency with human creativity by allowing Artificial Intelligence systems to analyze information and generate recommendations while keeping humans responsible for final decisions. Recent research in interactive AI and creative computing has demonstrated that collaborative systems can achieve better outcomes than either fully manual workflows or fully autonomous automation. In the context of video editing, HITL approaches enable AI systems to perform repetitive computational tasks such as footage analysis, synchronization, scene classification, and quality evaluation while allowing editors to accept, reject, or modify generated suggestions. However, existing HITL video editing studies primarily focus on individual tasks such as transcript-based editing, content retrieval, or simple recommendation mechanisms. Limited research has explored a complete Human-in-the-Loop workflow specifically designed for multi-footage editing, where AI assists with camera selection, scene organization, and timeline construction while preserving professional creative control.

Building upon these research advancements, SmartEdit introduces a Human-in-the-Loop AI framework specifically designed for intelligent multi-footage video editing. Unlike existing automatic editing approaches that attempt to generate complete videos independently, SmartEdit focuses on creating a collaborative editing environment where Artificial Intelligence functions as an intelligent assistant. The framework integrates multiple analysis capabilities, including multi-camera synchronization, duplicate scene detection, face recognition, event identification, audio-video quality assessment, and camera angle recommendation, into a unified workflow. Instead of relying on a single automated decision, SmartEdit generates explainable editing recommendations based on multiple visual and audio characteristics. These recommendations are presented through an editable timeline where users maintain complete authority over clip selection, sequence arrangement, transitions, and final creative decisions.

By combining computer vision-based footage analysis with Human-in-the-Loop interaction, SmartEdit addresses the major limitations identified in existing literature: the excessive manual effort of traditional editing systems, the lack of contextual understanding in rule-based automation, and the limited user control of fully autonomous AI editors. The proposed framework shifts the focus from replacing human editors toward augmenting their capabilities through intelligent assistance. Through automated multi-footage organization, quality-based clip recommendation, semantic event understanding, and interactive timeline refinement, SmartEdit establishes a practical and scalable approach for next-generation AI-assisted video editing systems suitable for professional content creation environments.

CHAPTER 3

ENVIRONMENTAL STUDY

This chapter describes the hardware and software environment required for the successful development and execution of the proposed SmartEdit framework. The system integrates Artificial Intelligence (AI), Large Language Models (LLMs), Computer Vision (CV), speech processing, and multimedia analysis to provide intelligent assistance for video editing. The proposed framework performs AI-based clip analysis, natural language understanding, content-aware editing recommendations, and Human-in-the-Loop interaction to simplify repetitive editing tasks while preserving human creative control. Therefore, an appropriate computing environment is required to efficiently execute AI models, process multimedia content, and support interactive editing operations.

The chapter presents the system configuration, hardware and software requirements, development environment, and the major technologies used to implement the SmartEdit framework. The chapter presents the system configuration, development environment, software specifications, and the major technologies used to implement the SmartEdit framework.

3.1 System Configuration

The SmartEdit framework requires a computing environment capable of processing high-resolution video clips, executing Artificial Intelligence models, and generating intelligent editing recommendations. The system configuration consists of hardware resources for multimedia processing and software components responsible for natural language understanding, video analysis, speech processing, and AI-assisted editing.

3.1.1 *Hardware Requirements*

The hardware requirements define the minimum computational resources necessary for processing video clips, executing Artificial Intelligence models, and performing multimedia analysis. Since video editing involves frame extraction, AI inference, and audio processing, sufficient processing capability, memory, graphics performance, and storage are required for efficient system operation.

The key hardware requirements are detailed as follows:

- **Processor:** Intel® Core™ i7 / AMD Ryzen™ 7 processor or higher, capable of handling parallel video processing and AI inference operations.
- **Memory:** Minimum 16 GB RAM recommended for simultaneous processing of multiple video streams, frame extraction, and machine learning model execution.
- **Graphics Processing Unit (GPU):** NVIDIA GPU with CUDA support (RTX series recommended)
- **Storage:** Minimum 100 GB free SSD storage for storing raw video footage, extracted frames, AI model files, temporary processing data, and generated output videos.
- **Display:** Full HD or higher resolution display for accurate video preview, timeline interaction, and editing interface visualization.

3.1.2 *Software Requirements*

The software environment specifies the operating system, programming language, frameworks, and libraries required for implementing the SmartEdit framework. These software components provide the necessary capabilities for Artificial Intelligence processing, Computer Vision, speech analysis, natural language understanding, and video editing.

The key software requirements are detailed as follows:

- **Operating System:** Microsoft Windows 11 / Ubuntu 22.04 LTS or later versions with support for GPU acceleration and multimedia processing.
- **Programming Language:** Python 3.10 or later, serving as the primary implementation language due to its extensive support for Artificial Intelligence and computer vision libraries.
- **Deep Learning Framework:** PyTorch or TensorFlow framework for executing trained AI models and performing computer vision inference.
- **Computer Vision Library:** OpenCV library for image processing, frame extraction, motion analysis, camera quality evaluation, and video manipulation.
- **Video Processing Engine:** FFmpeg framework for video decoding, encoding, synchronization, format conversion, and final video rendering.
- **Version Control System:** Git distributed version control system for source code management and collaborative development.

3.1.3 Software Requirements for Development

The development environment consists of programming tools, integrated development environments, Artificial Intelligence frameworks, and multimedia libraries required for implementing, testing, and maintaining the SmartEdit framework. These tools support software development, debugging, AI model integration, multimedia processing, and Human-in-the-Loop video editing.

The key development software requirements are detailed as follows:

- **Operating System:** Microsoft Windows 11 for application development and AI model execution.
- **Programming Language:** Python 3.10+ for implementing the AI pipeline, video analysis, and intelligent editing workflow.

- **Integrated Development Environment (IDE):** Visual Studio Code for application development, debugging, and project management.
- **Version Control:** Git and GitHub for source code management and project collaboration.
- **Artificial Intelligence Framework:** PyTorch for executing deep learning models and AI inference.
- **Computer Vision Library:** OpenCV for video frame processing, motion analysis, and clip quality assessment.
- **Object Detection Model:** YOLO for object and person detection.
- **Vision-Language Model:** CLIP for scene understanding and semantic analysis.
- **Speech Recognition Model:** Whisper for speech transcription and silence detection.
- **Audio Analysis Library:** Librosa for audio feature extraction and silence analysis.
- **Video Processing Library:** MoviePy for trimming, merging, and exporting edited video clips.
- **Large Language Model:** An LLM is used to understand user editing prompts, interpret editing intent, generate intelligent editing recommendations, and assist in creating the initial draft timeline.
- **Development Libraries and Frameworks:**
 - **OpenCV:** Used as the core computer vision library for video frame analysis, image processing, motion detection, camera angle comparison, object tracking, duplicate scene detection, and visual feature extraction from multiple video sources.

- **YOLOv8**: Used as an advanced deep learning-based object detection model for identifying humans, objects, activities, and important visual elements in video frames to assist AI-based clip selection and scene understanding.
- **MediaPipe**: Used for real-time human analysis tasks such as face detection, facial landmark extraction, pose estimation, and tracking important human features for selecting suitable footage.
- **PySceneDetect**: Used for automatic video scene segmentation by detecting shot boundaries and dividing long recordings into meaningful individual scenes for further AI analysis.
- **Librosa**: Used for audio signal processing, speech analysis, audio feature extraction, silence detection, beat analysis, and assisting video-audio synchronization between multiple recordings.
- **MoviePy**: Used as a video processing library for automated editing operations such as trimming unwanted sections, merging selected clips, applying basic transformations, and generating AI-edited video outputs.
- **Whisper Model**: Used for automatic speech recognition (ASR), converting audio into text, generating subtitles, and extracting semantic information from spoken content to support intelligent editing decisions.
- **PyTorch**: Used as the deep learning framework for implementing, training, and running AI models required for video analysis, feature extraction, and intelligent recommendation systems.
- **TensorFlow/Keras**: Used for developing and integrating machine learning models for classification tasks, content understanding, and future AI-based video editing improvements.
- **NumPy**: Used for numerical operations, matrix processing, image feature representation, and handling data generated during AI-based video analysis.

- **Pandas:** Used for managing, processing, and storing extracted video metadata, AI recommendations, timestamps, and editing information.
- **Scikit-learn:** Used for implementing machine learning algorithms for feature comparison, similarity measurement, classification, and ranking of video clips.
- **CUDA and cuDNN:** Used for GPU acceleration to improve the performance of deep learning models and enable faster processing of high-resolution video data using NVIDIA GPUs.
- **FFmpeg:** Used for low-level video and audio processing operations such as format conversion, encoding, decoding, frame extraction, and optimized media handling.
- **NLTK/SpaCy:** Used for natural language processing tasks such as analyzing user editing commands, extracting keywords from prompts, and supporting AI-based editing instructions.
- **React.js / Qt Framework:** Used for developing the user interface of the video editing environment, allowing users to review, modify, accept, or reject AI-generated editing suggestions.

3.2 *Software Specification*

The following section describes the major software components and technologies used in the SmartEdit framework. Each component performs a specific function within the intelligent video editing pipeline, including video input processing, AI-based analysis, recommendation generation, timeline management, and final video production.

3.2.1 *Python*

Python acts as the core orchestration language for the SmartEdit framework. It manages the complete AI-assisted editing workflow by

coordinating video input processing, computer vision analysis, machine learning model execution, metadata generation, and recommendation algorithms. The availability of extensive AI and multimedia libraries makes Python suitable for integrating different components of the intelligent editing pipeline.

3.2.2 *OpenCV*

OpenCV functions as the primary computer vision engine of SmartEdit. It provides essential capabilities for extracting video frames, analysing visual properties, detecting motion patterns, measuring sharpness, evaluating brightness, and identifying camera stability. These visual characteristics are used by the recommendation engine to determine the quality and suitability of individual clips.

3.2.3 *YOLOv8 Object Detection Model*

The YOLOv8 deep learning model is integrated into SmartEdit for real-time object and person detection. It enables the system to identify important subjects within video footage, such as people, vehicles, objects, and event-related elements. The detected information contributes to semantic understanding of each clip and assists in generating intelligent editing recommendations.

3.2.4 *MediaPipe Face Analysis*

MediaPipe provides efficient face detection and facial landmark analysis capabilities. Within SmartEdit, it is used to determine whether important subjects are visible, estimate face positioning, and evaluate the importance of individual camera angles. This information helps the system select visually meaningful clips during timeline generation.

3.2.5 FFmpeg Multimedia Processing Framework

FFmpeg acts as the underlying multimedia processing engine responsible for video decoding, encoding, format conversion, synchronization, and rendering operations. It enables SmartEdit to efficiently process multiple video formats and generate the final edited output.

3.2.6 AI Recommendation Engine

The recommendation engine represents the intelligent decision-support component of SmartEdit. Instead of automatically replacing the editor, it evaluates multiple parameters such as visual quality, stability, face visibility, audio quality, and scene importance to generate ranked clip recommendations. These recommendations are presented to the user through an editable timeline where the editor can accept, modify, or reject AI-generated suggestions.

3.2.7 Human-in-the-Loop Editing Interface

The Human-in-the-Loop interface provides interaction between the AI system and the editor. The interface displays recommended clips, detected events, camera selections, and timeline arrangements while allowing users to manually refine the generated output. This approach ensures that Artificial Intelligence improves productivity without limiting creative control.

3.2.8 Database and Metadata Management

SmartEdit maintains structured metadata for every analyzed video clip. Information such as detected objects, scene category, quality score, audio properties, and AI recommendations are stored and retrieved during editing operations. This metadata layer enables

efficient searching, ranking, and organization of large collections of multimedia content.

3.2.9 Visualization and Timeline Engine

The visualization engine provides an interactive editing environment where users can view AI-generated recommendations and modify the final sequence. It represents video clips, scene boundaries, camera sources, and editing decisions through an intuitive timeline interface. This component transforms AI analysis results into a practical editing workflow suitable for both professional and beginner users.

CHAPTER 4

SYSTEM ANALYSIS

The practice of performing system analysis in a technical development scenario focuses on identifying the most effective approach to solving a specific problem through systematic evaluation of existing challenges, user requirements, and technological possibilities. In the context of SmartEdit, the analysis focuses on addressing the difficulties faced by content creators and professional video editors in managing, organizing, and processing large volumes of multi-camera footage. Before initiating the development of such an intelligent video editing framework, project stakeholders, including researchers and student developers, define the project objectives, analyze the existing editing workflow, and prepare a structured proposal for evaluation and approval.

A structured system investigation in this context involves studying the limitations of conventional video editing approaches and evaluating the potential of Artificial Intelligence, Computer Vision, and multimedia processing techniques to improve the editing workflow. Traditional editing platforms provide extensive creative control but require significant manual effort for tasks such as footage review, synchronization, clip selection, and timeline arrangement. On the other hand, fully automated AI editing solutions reduce manual workload but often lack user control and fail to understand individual creative preferences. System analysis helps identify this gap and defines the requirement for a Human-in-the-Loop (HITL) framework where

AI assists editors by generating intelligent recommendations while allowing human users to make final decisions.

The primary objective of system analysis for SmartEdit is to decompose the video editing problem into multiple functional components, including multi-footage input management, video synchronization, scene understanding, quality assessment, camera angle recommendation, timeline generation, and user-controlled refinement. Each component is analyzed to determine the required Artificial Intelligence models, Computer Vision algorithms, multimedia processing techniques, and user interaction mechanisms needed for effective implementation.

Furthermore, system analysis ensures that the proposed framework maintains a balance between automation and creative freedom. The system must not completely replace the editor but should act as an intelligent assistant capable of reducing repetitive tasks and improving productivity. Through detailed analysis of functional and non-functional requirements, SmartEdit establishes a foundation for developing an efficient, scalable, and user-friendly AI-assisted video editing environment capable of supporting professional and amateur content creators.

4.1 Requirements Analysis

Requirement analysis is a crucial phase in the software development life cycle that focuses on identifying, documenting, and analyzing the functional and technical requirements of a system before the design and implementation stages. For the proposed SmartEdit framework, which aims to provide a Human-in-the-Loop (HITL) Artificial Intelligence-assisted multi-footage video editing environment, this phase plays an essential role in defining the system scope, expected behavior, processing workflow, and interaction between artificial intelligence

modules and human editors.

The primary requirement of the system is to support multiple video inputs captured from different recording devices such as smartphones, DSLR cameras, drones, and action cameras. The system must be capable of accepting heterogeneous video sources containing different resolutions, frame rates, formats, and recording conditions. These input videos should be processed and organized into a unified workspace where further AI-based analysis and editing operations can be performed. The preprocessing stage should extract essential metadata such as video duration, resolution, frame rate, timestamps, and audio information to support subsequent analysis modules.

Another key requirement of SmartEdit is the integration of Artificial Intelligence and Computer Vision techniques for intelligent video understanding. The system must analyze uploaded footage to extract meaningful visual and audio features, including scene information, object presence, facial characteristics, motion patterns, camera stability, brightness, sharpness, and audio quality. These extracted features act as contextual information that enables the system to understand the importance, quality, and suitability of individual video segments for final editing.

The system must provide automatic multi-camera synchronization capabilities by analyzing temporal relationships between different recordings of the same event. Audio signals, timestamps, and visual similarity patterns should be evaluated to align multiple video streams accurately. This eliminates the requirement for editors to manually synchronize recordings and improves efficiency when handling large-scale event footage.

Another important requirement is the development of an AI-based clip analysis and recommendation module. The system should evaluate every video segment based on multiple quality parameters such as visual clarity, subject visibility, camera stability, composition, motion

smoothness, and event relevance. Using these parameters, SmartEdit should rank available clips and recommend the most appropriate shots and camera angles for different parts of the final video sequence.

A significant requirement of the proposed system is the implementation of a Human-in-the-Loop editing approach. Unlike fully automated AI editing systems, SmartEdit must allow users to maintain complete creative control over the editing process. AI-generated suggestions should be presented through an interactive interface where editors can review, modify, accept, or reject recommendations. This collaborative approach ensures that artificial intelligence assists the creative process rather than replacing human decision-making.

The system must also support automatic scene analysis and duplicate footage detection. Large-scale recordings often contain repeated scenes, unused clips, and low-quality segments that increase editing complexity. Therefore, SmartEdit should identify visually similar scenes, detect unnecessary footage, and assist users in selecting meaningful content for the final composition.

Furthermore, the system requires an intelligent timeline generation mechanism that automatically organizes recommended clips into an editable sequence. The generated timeline should allow users to perform conventional editing operations such as trimming, rearranging clips, changing camera selections, applying transitions, adjusting audio levels, and exporting the completed video. The timeline should provide flexibility similar to professional editing software while reducing the manual effort required for initial arrangement.

In addition to functional requirements, several non-functional requirements must be satisfied. The system should provide efficient processing performance while handling multiple high-resolution video files. It must maintain scalability to support different event types and varying amounts of input footage. The AI recommendation process should be reliable, consistent, and understandable to users through

explainable feedback. The user interface should be simple, accessible, and suitable for both professional editors and beginners.

Security and privacy are also important requirements because the system processes user-generated multimedia content. Uploaded videos must be handled securely, and appropriate mechanisms should be implemented to prevent unauthorized access or misuse of personal media files. The architecture should be modular and extensible, allowing future integration of advanced AI models, additional editing features, and improved multimedia processing techniques.

This requirement analysis ensures that the SmartEdit framework is aligned with its intended objectives by combining Artificial Intelligence automation with human creative control. It provides a strong foundation for the subsequent phases of system design, implementation, testing, and evaluation while addressing the limitations of traditional manual editing workflows and fully automated AI editing systems.

4.2 Existing System

Existing video editing systems are primarily designed around manual editing workflows where users are responsible for organizing footage, selecting appropriate clips, synchronizing multiple recordings, applying transitions, and creating the final sequence. Professional editing platforms such as Adobe Premiere Pro, DaVinci Resolve, and Final Cut Pro provide powerful creative tools and complete control over the editing process. However, these systems require significant technical knowledge, editing experience, and manual effort, especially when handling large volumes of multi-camera footage captured from smartphones, DSLR cameras, drones, and action cameras.

Traditional video editing applications mainly focus on providing a comprehensive set of editing tools rather than intelligent decision-making capabilities. Tasks such as identifying the best shots, removing

duplicate scenes, selecting suitable camera angles, synchronizing audio and video streams, and maintaining consistent visual quality must often be performed manually by editors. This becomes highly time-consuming in applications such as wedding cinematography, event coverage, documentaries, and social media content production, where hundreds of recorded clips need to be reviewed before final editing.

Recent developments in Artificial Intelligence and Computer Vision have introduced automated video editing solutions that attempt to reduce manual workload. These systems utilize techniques such as scene detection, face recognition, object tracking, audio analysis, and machine learning-based recommendations to automate specific editing operations. However, many existing AI-based editing systems operate as fully automated pipelines, where the generated output is directly produced without allowing sufficient user interaction or creative control. As a result, automated decisions may not always match the editor's artistic intention, storytelling requirements, or preferred editing style.

Another significant limitation of existing systems is the lack of a collaborative Human-in-the-Loop editing approach. Most AI-powered solutions provide recommendations or automatically generate videos but do not allow users to continuously review, modify, accept, or reject individual AI decisions. This creates a gap between automation efficiency and professional creative control. Editors require intelligent assistance that can reduce repetitive tasks while still keeping them involved in the decision-making process.

Some of the widely used existing systems include:

- **Adobe Premiere Pro:** Adobe Premiere Pro is a professional non-linear video editing platform widely used in film production, broadcasting, and online content creation. It provides advanced features such as multi-track timeline editing, color grading, audio processing, motion graphics integration, and multi-camera

editing support. Although it offers powerful manual control, most editing decisions depend entirely on the user. The editor must manually inspect footage, synchronize multiple camera recordings, select suitable shots, and arrange clips according to the desired storytelling flow. While recent versions include AI-powered features such as automatic reframing and speech enhancement, they mainly focus on individual editing tasks rather than complete intelligent footage analysis and camera selection. The system does not provide a complete AI-assisted workflow where recommendations are generated, explained, and refined through continuous human interaction.

- **DaVinci Resolve:** DaVinci Resolve is an advanced video editing and post-production software widely recognized for professional color correction, visual effects, audio processing, and editing capabilities. The platform includes AI-powered tools through its Neural Engine, such as facial recognition, object tracking, noise reduction, and automatic scene detection. However, these features mainly improve individual editing operations rather than understanding the overall context of multiple footage recordings. The system does not automatically analyze all available camera streams of the same event, evaluate footage quality, and recommend the optimal sequence based on cinematic requirements. Additionally, professional-level features require considerable learning experience, making the system difficult for beginners.
- **AI-Based Automatic Video Editors:** Several AI-powered video editing platforms attempt to simplify content creation by automatically generating videos from raw footage. These systems analyze visual and audio information using machine learning techniques and perform operations such as automatic trimming, highlight extraction, background music selection,

and caption generation. While these approaches significantly reduce editing time, they generally follow predefined models or learned patterns and provide limited customization options. The generated edits may fail to represent the creator's personal style, emotional storytelling requirements, or event-specific importance. Furthermore, many systems provide limited transparency regarding why a particular clip or transition was selected.

- **Multimodal AI Video Editing Systems:** Recent research has explored multimodal interfaces where users can interact with video editors using natural language prompts, sketches, or voice commands. These systems improve accessibility by allowing users to describe desired modifications instead of manually operating complex editing tools. However, most existing multimodal editors focus on direct command execution rather than comprehensive footage understanding. They often lack advanced multi-camera analysis, quality evaluation, personalized editing recommendations, and a feedback mechanism where the system learns from user corrections.

Although existing video editing platforms provide powerful creative capabilities and AI-based automation features, they still lack an integrated framework that combines intelligent footage analysis, automatic recommendation generation, and continuous human supervision. Current solutions either depend heavily on manual editing effort or provide fully automated outputs with limited user control. This highlights the need for a Human-in-the-Loop AI-assisted editing framework that can reduce repetitive editing tasks while preserving the creative authority of the editor.

4.3 *Limitations of Existing System*

Although existing video editing platforms provide powerful editing capabilities and AI-assisted features, they still have several limitations when dealing with complex multi-footage editing workflows. Current solutions either depend heavily on manual editing operations or provide automated outputs with limited user involvement. The major drawbacks of existing systems are as follows:

- **High Dependency on Manual Footage Organization:** Traditional video editing applications require editors to manually review, categorize, synchronize, and arrange large volumes of raw footage before beginning the actual editing process. When multiple recordings of the same event are captured from different cameras, such as smartphones, DSLR cameras, drones, and action cameras, identifying the most suitable clips and camera angles becomes highly time-consuming. Existing systems lack intelligent mechanisms to automatically understand the relationship between different footage sources and organize them efficiently.
- **Limited Context Understanding of Video Content:** Most existing editing systems perform operations based on predefined rules or isolated AI functions such as scene detection, object tracking, or face recognition. These approaches often fail to understand the complete context of a video sequence, including the importance of specific events, emotional moments, speaker interactions, or storytelling requirements. As a result, automatically generated edits may include technically correct clips but fail to maintain the intended narrative flow.
- **Lack of Intelligent Multi-Camera Selection:** Existing multi-camera editing solutions are generally designed for specific domains such as sports broadcasting, surveillance, or professional studio production. These systems usually depend on fixed

camera-selection rules and require controlled environments. They cannot effectively analyze informal multi-camera recordings such as weddings, interviews, educational videos, and social media content where camera quality, positioning, and recording conditions vary significantly.

- **Limited Human-AI Collaboration:** Many existing AI-powered video editors focus on complete automation, where the system independently generates the final output without providing sufficient control to the user. Such approaches may produce results that do not match the creator’s artistic vision or personal editing style. The absence of a Human-in-the-Loop workflow prevents editors from reviewing, modifying, accepting, or rejecting individual AI-generated recommendations during the editing process.
- **Absence of Comprehensive Footage Quality Evaluation:** Current editing software provides separate tools for checking technical aspects such as stabilization, brightness, audio quality, or sharpness. However, these features are usually independent and do not provide a unified quality assessment mechanism for comparing multiple clips. Editors must manually evaluate factors such as camera stability, focus quality, face visibility, audio clarity, and visual composition before selecting the best footage.
- **Limited Explainability of AI-Based Editing Decisions:** Existing AI editing systems often generate automatic cuts, transitions, or recommendations without explaining the reasoning behind their decisions. Users cannot easily understand why a particular clip was selected, why another clip was rejected, or how the system determined the importance of a scene. This lack of transparency reduces user trust and limits the ability of editors to effectively collaborate with AI systems.
- **Lack of Personalized Editing Adaptation:** Most existing video

editing tools provide generic automation features that apply the same editing patterns to different users. They do not learn from individual editing preferences, previous corrections, or creative styles. As a result, the generated output may not reflect the unique requirements of different creators or production environments.

- **Fragmented AI Editing Workflow:** Current solutions often provide AI capabilities as separate independent features, such as automatic captions, noise reduction, color enhancement, or object detection. These isolated tools do not form a complete intelligent editing pipeline that connects footage analysis, clip recommendation, timeline arrangement, and final refinement. This fragmentation increases workflow complexity and prevents seamless AI-assisted editing.

4.4 Proposed System

The proposed system, SmartEdit, is designed to provide a Human-in-the-Loop (HITL) AI-assisted multi-footage video editing framework by integrating Artificial Intelligence, Computer Vision, multimedia processing, and interactive editing technologies. The system aims to reduce the complexity and time consumption involved in professional video editing by automating repetitive tasks while preserving complete creative control for the user. Unlike conventional video editing applications that require extensive manual effort or fully automated AI editors that provide limited customization, SmartEdit establishes a collaborative workflow where artificial intelligence assists editors through intelligent recommendations and users maintain final decision-making authority.

The system accepts multiple video recordings captured from different devices such as smartphones, DSLR cameras, drones, and action cameras representing the same event or scene. Instead of treating

each video independently, SmartEdit analyzes the complete collection of footage to understand visual content, temporal relationships, audio characteristics, and technical quality parameters. The framework performs automatic video synchronization, duplicate scene detection, camera quality evaluation, face and object recognition, motion analysis, and important event identification to generate meaningful editing recommendations.

The analyzed video information is transformed into a structured editing representation containing clip metadata, quality scores, detected events, and recommended camera selections. These intelligent recommendations are presented through an interactive timeline interface where users can review AI-generated suggestions, modify clip arrangements, select alternative camera angles, adjust transitions, and apply manual refinements. The final editing decision always remains under human control, ensuring that creative preferences and storytelling requirements are preserved.

The proposed SmartEdit architecture consists of the following core modules:

- **Multi-Footage Input and Preprocessing Module:** This module accepts multiple video files recorded from different cameras and performs initial preprocessing operations. It extracts essential metadata such as resolution, frame rate, duration, timestamps, and audio information. The module prepares the raw footage for further analysis by converting different input formats into a unified processing structure. It also performs basic validation to ensure that uploaded videos are suitable for analysis.
- **Video Synchronization and Alignment Module:** This module automatically synchronizes multiple recordings of the same event using visual and audio-based matching techniques. Audio waveform comparison, timestamp analysis, and visual feature

matching are used to identify corresponding moments across different camera streams. By generating synchronized timelines, the system eliminates the manual effort required for aligning multi-camera footage.

- **Computer Vision-Based Footage Analysis Module:** This module performs intelligent analysis of video content using computer vision techniques. It identifies important visual characteristics such as scene changes, human faces, objects, motion intensity, camera stability, brightness, and sharpness. The module calculates quality scores for each clip based on technical and semantic factors, allowing the system to determine the most suitable footage for final editing.
- **AI-Based Clip Recommendation and Camera Selection Module:** This module analyzes extracted video features and generates intelligent recommendations for selecting the best clips and camera angles. The system evaluates multiple factors including subject visibility, emotional importance, visual quality, movement, and scene relevance. Instead of automatically replacing the editor's decision, the module provides ranked suggestions that users can accept, reject, or modify according to their creative requirements.
- **Event Understanding and Scene Classification Module:** This module identifies important events and classifies different video segments based on their content. Using artificial intelligence models, the system can recognize activities such as speeches, performances, interactions, highlights, and significant moments. This enables SmartEdit to prioritize meaningful clips and construct a more engaging video sequence.
- **Human-in-the-Loop Interactive Editing Module:** The interactive editing module provides a user-controlled environment where AI recommendations are displayed within an editable timeline interface. Users can review suggested clips, rearrange sequences,

change camera selections, trim unwanted sections, modify transitions, and provide feedback to the system. This continuous interaction creates a collaborative editing workflow where AI supports human creativity rather than replacing it.

- **AI Feedback and Personalization Module:** This module collects user interactions such as accepted recommendations, rejected clips, preferred camera angles, and editing modifications. These feedback signals can be used to improve future recommendations and adapt the editing process according to individual user preferences. The system gradually learns the editing style and requirements of different creators.
- **Final Rendering and Export Module:** After completing the editing process, the final timeline configuration is processed by the rendering module. This module combines selected clips, applied transitions, audio adjustments, and user modifications to generate the final output video. The system supports exporting the edited content in commonly used video formats suitable for professional production and social media platforms.

4.5 Advantages of Proposed System

The proposed SmartEdit framework provides several significant advantages by combining Artificial Intelligence, Computer Vision, multimedia processing, and Human-in-the-Loop interaction into a unified video editing environment. By integrating automated footage analysis with user-controlled editing decisions, the system improves editing efficiency while preserving the creative freedom required for professional content production.

- **Intelligent Multi-Footage Analysis:** Unlike traditional editing systems where users manually inspect and organize hundreds of video clips, SmartEdit automatically analyzes multiple recordings

from different cameras and identifies relationships between them. The system evaluates visual quality, audio information, scene content, and camera characteristics to reduce the time required for footage management.

- **Automated Clip Selection and Camera Recommendation:** The framework intelligently evaluates available clips and recommends the most suitable footage based on factors such as sharpness, stability, face visibility, motion quality, and event importance. This helps editors quickly identify high-quality shots and appropriate camera angles without manually reviewing every recording.
- **Human-in-the-Loop Creative Control:** Unlike fully automated AI video editors, SmartEdit keeps the editor involved throughout the editing process. Users can review, accept, reject, or modify AI-generated recommendations, ensuring that the final output reflects the creator's artistic vision, storytelling style, and personal preferences.
- **Reduced Editing Time and Manual Effort:** The system automates repetitive editing tasks such as footage synchronization, duplicate scene identification, quality evaluation, and initial timeline arrangement. By reducing these time-consuming operations, editors can focus more on creative decisions rather than technical preparation.
- **Context-Aware Video Understanding:** SmartEdit does not rely only on basic video properties but analyzes the semantic meaning of footage using Artificial Intelligence and Computer Vision techniques. The system can identify important events, human interactions, and visually significant moments, enabling more meaningful clip recommendations.
- **Automatic Multi-Camera Synchronization:** The proposed framework eliminates the difficulty of manually aligning multiple recordings by automatically synchronizing video and audio

streams. This feature is especially beneficial for wedding videos, interviews, sports events, and documentary production where multiple cameras are commonly used.

- **Quality-Based Footage Ranking:** The system provides an intelligent ranking mechanism that evaluates clips according to technical and visual quality parameters. Factors such as camera stability, lighting conditions, audio clarity, focus accuracy, and subject visibility are considered to identify the best available footage.
- **Interactive AI-Assisted Editing Timeline:** SmartEdit provides an editable timeline environment where AI recommendations are integrated directly into the editing workflow. Users can modify clip order, replace camera angles, adjust transitions, and perform manual refinements without leaving the AI-assisted environment.
- **Personalized Editing Experience:** The framework can utilize user feedback and editing decisions to understand individual preferences. By learning from accepted or rejected recommendations, SmartEdit can gradually adapt its suggestions according to the creator's editing style and workflow requirements.
- **Applicable Across Multiple Domains:** The proposed system is not restricted to a single application area. Its intelligent multi-footage editing capability can support various domains including wedding cinematography, event coverage, interviews, educational content, documentaries, sports production, and social media video creation.

4.6 Functional Requirements

The proposed SmartEdit system is designed to perform intelligent multi-footage video analysis, AI-assisted clip recommendation, automatic synchronization, and Human-in-the-Loop video editing

by leveraging Artificial Intelligence, Computer Vision, multimedia processing techniques, and interactive timeline management. The functional requirements define the expected behavior, user interactions, processing capabilities, and internal operations required to achieve an efficient AI-assisted video editing workflow.

The system shall accept multiple video recordings captured from different sources such as smartphones, DSLR cameras, drones, and action cameras. These input videos represent the same event or scene and are processed through a unified analysis pipeline. The framework shall extract meaningful information from each video, including metadata, audio characteristics, visual features, scene information, and quality parameters.

The system shall perform automatic video synchronization using audio and visual similarity analysis to align multiple camera recordings accurately. It shall analyze each video segment using computer vision techniques to identify important events, detect faces and objects, evaluate camera stability, measure visual quality, and determine scene relevance.

The system shall generate intelligent editing recommendations by ranking clips based on multiple factors such as technical quality, subject visibility, motion characteristics, audio clarity, and event importance. These recommendations shall be presented to users through an interactive timeline interface where editors can review, modify, accept, or reject AI-generated suggestions.

Furthermore, the system shall support Human-in-the-Loop interaction by allowing users to manually refine AI decisions, rearrange clips, select alternative camera angles, adjust transitions, and customize the final sequence according to their creative requirements. The system shall maintain user feedback information to improve future recommendations and adapt to individual editing preferences.

The framework shall provide final video rendering and export capabilities after completing the editing process. It shall combine selected clips, applied effects, transitions, and audio modifications to generate the final output video in commonly used formats. In addition to functional operations, the system shall ensure efficient processing, reliable analysis, user-friendly interaction, and transparent AI-assisted decision-making throughout the complete editing workflow.

4.6.1 Multi-Footage Video Input and Upload

The system shall provide a dedicated interface for users to upload multiple video recordings captured from different devices, including smartphones, DSLR cameras, drones, action cameras, and professional video cameras. The uploaded footage may contain different resolutions, frame rates, aspect ratios, encoding formats, and audio characteristics. The system shall support commonly used multimedia file formats such as MP4, MOV, AVI, and MKV to ensure compatibility with a wide range of recording devices.

Before initiating the AI-based analysis pipeline, the system shall validate all uploaded video files to ensure that they are accessible, properly encoded, and contain valid multimedia streams. The preprocessing module shall verify essential metadata information such as video duration, resolution, frame rate, codec type, audio availability, and file integrity. If unsupported file formats, corrupted files, or incomplete uploads are detected, the system shall generate appropriate error messages and request valid input files from the user.

The system shall allow users to upload multiple recordings of the same event simultaneously and organize them into a single project workspace. Each uploaded video shall be assigned a unique identifier and stored with its corresponding metadata information to maintain proper tracking throughout the editing workflow. The system shall preserve the original source footage without modification, ensuring

that all AI-generated operations are performed on extracted analysis data rather than directly altering user files.

Additionally, the system shall provide options for users to add, remove, rename, and manage uploaded footage before beginning the automated analysis process. The interface shall display basic information about each video source, including file name, duration, resolution, recording device information, and preview thumbnails. This functionality enables users to verify their selected media assets and maintain complete control over the input data before AI processing begins.

4.6.2 Video Preprocessing and Metadata Extraction

The system shall perform an automated preprocessing stage after successful video upload to prepare the input footage for AI-based analysis and editing operations. This module shall process multiple video streams captured from different devices and normalize their technical properties without modifying the original source files. The preprocessing pipeline shall extract important multimedia characteristics including resolution, frame rate, duration, codec information, timestamp details, orientation, and audio channel properties.

The system shall analyze the extracted metadata to identify variations between different recordings. Differences in frame rates, camera settings, encoding formats, and resolution shall be detected and stored as reference information for later synchronization and editing operations. The preprocessing module shall also generate optimized preview representations and thumbnail frames to enable faster browsing and timeline interaction.

Additionally, the system shall perform video frame extraction and feature preparation required for computer vision algorithms. Important

visual characteristics such as motion patterns, color information, scene boundaries, and object details shall be extracted from selected frames. The generated analysis data shall be stored separately from the original media files to maintain data integrity and improve processing efficiency during subsequent AI operations.

4.6.3 AI-Based Video Synchronization

The system shall automatically synchronize multiple recordings of the same event by analyzing visual and audio similarities between different video sources. Since footage captured from different cameras may contain different starting points and recording durations, the synchronization module shall identify common reference points and align all available recordings into a unified timeline.

The system shall utilize audio waveform analysis, timestamp comparison, motion pattern matching, and visual feature extraction techniques to determine accurate synchronization positions. In situations where multiple cameras capture the same action from different angles, the system shall identify corresponding frames and calculate the temporal relationship between each recording.

The synchronized output shall allow users to view and edit multiple camera perspectives simultaneously without manually matching clips. The system shall maintain synchronization information throughout the editing process, ensuring that camera switching, trimming, and AI-generated recommendations remain temporally accurate.

4.6.4 Scene Detection and Duplicate Removal

The system shall provide an automated scene analysis mechanism to identify meaningful segments within large collections of uploaded footage. The scene detection module shall analyze visual changes,

camera movements, transition points, and content variations to divide continuous videos into logically separated scenes.

The system shall detect duplicate or highly similar video segments captured from multiple cameras using computer vision-based feature comparison techniques. Repeated frames, unnecessary recordings, accidental duplicates, and low-value segments shall be identified and marked for removal or review.

The detected scenes shall be categorized based on their visual importance and stored as independent editing units. This process reduces the amount of footage that editors need to manually review and helps in creating a more efficient content selection workflow.

4.6.5 Audio and Video Quality Analysis

The system shall evaluate the technical and aesthetic quality of uploaded video footage using AI-based audio and visual analysis techniques. The quality analysis module shall examine parameters such as image sharpness, brightness level, exposure, camera stability, noise level, motion blur, and overall visual clarity.

The audio analysis component shall evaluate recorded sound quality by measuring background noise, volume consistency, speech clarity, and unwanted disturbances. The system shall identify clips containing poor audio conditions, excessive noise, or unclear speech and provide appropriate recommendations.

Based on the quality assessment results, each video segment shall receive a quality score representing its suitability for final editing. These quality measurements shall assist the recommendation engine in selecting the most appropriate clips for the final composition.

4.6.6 Face Recognition and Important Event Detection

The system shall integrate computer vision-based face recognition techniques to identify important individuals and significant events within the uploaded footage. The face analysis module shall detect and track human faces across multiple camera recordings and associate similar appearances throughout different scenes.

The system shall allow users to mark important persons, such as bride, groom, speakers, performers, or interview subjects, and prioritize footage containing these individuals. The recognition system shall improve clip selection by identifying moments where important subjects are visible, active, or interacting.

Furthermore, the system shall detect important events based on visual and audio patterns. Events such as emotional reactions, applause, speeches, celebrations, sports actions, or significant movements shall be automatically identified and highlighted for user review.

4.6.7 AI-Based Clip Recommendation

The system shall provide intelligent clip recommendation capabilities by analyzing multiple factors including video quality, subject importance, scene relevance, motion characteristics, and audio quality. The recommendation engine shall rank available video segments according to their suitability for inclusion in the final edited output.

The AI model shall evaluate different camera angles and automatically recommend the most effective viewpoint for each scene. For example, during event coverage, the system may prioritize a close-up camera angle for important moments while selecting wider shots for environmental context.

The generated recommendations shall be presented to the user through an interactive editing interface. Users shall have complete

authority to accept, modify, or reject AI-generated suggestions based on their creative requirements.

4.6.8 Human-in-the-Loop Timeline Editing

The system shall provide an interactive timeline editing environment where users can manually control and refine AI-generated editing decisions. Unlike fully automated editing systems, the proposed framework shall maintain continuous human involvement throughout the editing workflow.

The timeline interface shall allow users to arrange recommended clips, trim unnecessary sections, adjust clip duration, change camera angles, modify sequence order, and apply creative decisions manually. The system shall display AI suggestions alongside traditional editing controls to provide a collaborative editing experience.

All modifications performed by the user shall be tracked within the project workspace, allowing editors to experiment with different arrangements while maintaining complete control over the final video structure.

4.6.9 AI-Assisted Editing Suggestions

The system shall provide intelligent editing assistance by generating context-aware suggestions during the editing process. The AI module shall analyze the current timeline structure and recommend improvements related to clip ordering, camera switching, transition placement, and pacing.

The system shall suggest suitable edits based on the detected content type, including cinematic sequences, interview structures, event highlights, educational content, and social media formats. It shall recommend appropriate transitions, highlight important moments, and identify sections requiring improvement.

All AI-generated suggestions shall be presented as optional recommendations rather than automatic modifications. Users shall be able to review, accept, reject, or manually adjust each suggestion to preserve creative freedom.

4.6.10 Video Preview and Export

The system shall provide a real-time preview environment where users can review the edited timeline before generating the final output. The preview module shall combine selected clips, applied transitions, audio adjustments, and AI-assisted modifications to display the expected final result.

The export module shall support rendering the completed project into commonly used video formats such as MP4 and MOV. Users shall be able to configure export parameters including resolution, frame rate, video quality, and compression settings according to their requirements.

Before final rendering, the system shall validate the project timeline to ensure that all media references, synchronization data, and editing operations are correctly processed. The exported video shall maintain the selected quality settings while preserving synchronization between audio and video components.

4.6.11 Error Handling

The system shall provide comprehensive error handling mechanisms to manage failures during video upload, preprocessing, AI analysis, editing operations, and export procedures. The application shall detect invalid files, unsupported formats, corrupted media, insufficient system resources, and processing failures.

When an error occurs, the system shall provide meaningful messages describing the problem and possible corrective actions. The system

shall prevent data loss by maintaining project states and automatically preserving previously completed operations.

Additionally, the system shall handle AI processing failures, network interruptions, and unexpected exceptions without terminating the complete editing workflow. All errors shall be logged appropriately for debugging and future system improvement while ensuring a smooth user experience.

4.7 Non-Functional Requirements

The non-functional requirements define the quality attributes of the SmartEdit system. These requirements ensure that the system provides efficient performance, reliable AI-based recommendations, secure media processing, and a user-friendly editing experience.

4.7.1 Performance

The system shall perform video analysis, synchronization, scene detection, and AI-based recommendations efficiently. The processing pipeline should provide faster analysis by using optimized multimedia processing techniques and avoid unnecessary delays during editing operations.

4.7.2 Usability

The system shall provide an intuitive and user-friendly interface that allows both professional editors and beginners to easily manage footage, review AI suggestions, and modify the final timeline. The workflow should reduce complexity while maintaining creative control.

4.7.3 Accuracy and Reliability

The system shall provide accurate video synchronization, scene identification, quality analysis, and clip recommendations. AI-generated suggestions should be reliable and validated before being presented to users for final decisions.

4.7.4 Security and Privacy

The system shall protect user-uploaded video content and ensure that personal media files are securely processed. User data shall not be accessed or modified without authorization, maintaining privacy throughout the editing process.

4.7.5 Scalability and Extensibility

The system shall support increasing numbers of video files, higher-resolution footage, and additional AI features in the future. The modular architecture should allow integration of new models and editing capabilities without major changes.

4.7.6 Explainability and User Control

The system shall provide explanations for AI-generated recommendations, allowing users to understand why specific clips or edits are suggested. All AI decisions shall remain under human control, enabling users to accept, modify, or reject suggestions.

4.8 Feasibility Study

A feasibility study evaluates the practicality of implementing the SmartEdit framework by analyzing technical, operational, and economic factors.

4.8.1 Technical Feasibility

The system is technically feasible due to the availability of advanced AI models, computer vision libraries, and multimedia processing frameworks. Technologies such as Python, OpenCV, deep learning models, and video processing tools provide sufficient support for implementation.

4.8.2 Operational Feasibility

The system is operationally feasible as it provides a simple AI-assisted editing workflow. Users can upload footage, review recommendations, modify the timeline, and export videos without requiring advanced technical knowledge.

4.8.3 Economical Feasibility

The system is economically feasible because it can be developed using open-source technologies and existing AI frameworks. It reduces manual editing time and improves productivity, making it cost-effective for content creators and professional editors.

BIBLIOGRAPHY

- [1] T. Tilekbay, A. Kim, D. E. Park, and J. A. Landay, "ExpressEdit: A Multimodal System for Video Editing with Natural Language and Sketching," in *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024.
- [2] Z. Chen, K. Liang, H. Wang, *et al.*, "EditBoard: A Comprehensive Evaluation Benchmark for Text-based Video Editing," *arXiv preprint arXiv:2402.09257*, 2025.
- [3] K. Wang, Z. Wang, Z. Yuan, *et al.*, "LAVE: An LLM-powered Agent for Video Editing," *arXiv preprint arXiv:2402.19131*, 2024.
- [4] K. Shekar, K. Kasi Viswanadh, V. Srujan Varma, K. Veda Sri, S. Aditya, and S. Sahoo, "AI-Based Intelligent Video Editing System," in *2025 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC)*, 2025.
- [5] S. Tu, Q. Dai, Z. Zhang, S. Xie, Z.-Q. Cheng, C. Luo, X. Han, Z. Wu, and Y.-G. Jiang, "Paper Title," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [6] B. Li, Y. Wu, Y. Lu, J. Yu, L. Tang, J. Cao, W. Zhu, Y. Sun, J. Wu, and W. Zhu, "VEU-Bench: Towards Comprehensive Understanding of Video Editing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [7] V. Biyyala, B. C. Kathuria, J. Li, and Y. Zhang, "SST-EM: Advanced Metrics for Evaluating Semantic, Spatial and Temporal Aspects in Video Editing," *arXiv preprint*, 2025.

- [8] E. Jones and J. Black, "Integrating AI and User Feedback in Video Editing Software," *IEEE Access*, vol. 12, pp. 10234–10250, 2024.